

January -May 2026 Session

Programme	Bachelor of Computer Applications (BCA) Hons	
Course Code: CS1280	Course Name: Python Programming for Data Analytics	
Semester: II	Area: Major Computer Science Core	SEE Type: Practical (30 Marks)
Level:	Credits: 3(1: 0: 4)	Contact Hours: 75 Hours

Course Faculty:

Sr. No.	Section	Course Faculty Name	Contact: Email	Mobile
1	A	Dr. Mahadev Patil	mahadevkalyanraop@rvu.edu.in	99752 69569
2	B	Prof. Mrs. K Thenmozhi	kthenmozhi@rvu.edu.in	98405 59146

Lesson Plan:

Session	Module No.	Topic	Pedagogy /Activities	Reference for Study
1	1	Introduction to Python: history, features, environment setup	Lecture: Overview of Python, uses, installing Python/Jupyter.	Zelle Ch.1
2	1	Basic Syntax & Variables	Practical: Write first script, variables, expressions.	Zelle Ch.2
3, 4	1	Basic Programming (Lab Program 2A–C)	Practical: Simple calculator, temperature converter, largest among 3 numbers.	Lab Manual 2
5	1	Control Structures: if, elif, else	Lecture: Conditional logic with examples.	Zelle Ch.3
6,7	1	Conditional Programs	Practical: Decision-making programs.	Lab Manual
8	1	Loops: for, while, nested loops	Practical: Pattern programs, counters.	Zelle Ch.4
9	1	Functions & Modules	Lecture: Defining functions, parameters, return values.	Zelle Ch.5
10, 11	1	Functions & Recursion (Lab Program 3A–B)	Practical: Factorial recursion, word count.	Lab Manual 3
12	1	Integrated Problem Solving	Practical: Combine loops, conditions, functions.	Instructor Notes
13	1	Revision		
14	2	Data Structures Overview	Lecture: Lists, tuples, sets, dictionaries.	Zelle Ch.7

Session	Module No.	Topic	Pedagogy /Activities	Reference for Study
15, 16	2	Lists (Lab Program 4A)	Practical: Insert, delete, search operations.	Lab Manual 4
17, 18	2	Dictionaries (Lab Program 4B)	Practical: Merge dictionaries, comprehension.	Lab Manual
19	2	Arrays & 2D Arrays	Lecture: 1D/2D arrays, nested loops.	Zelle Ch.8
20	2	Array Operations	Practical: 2D array exercises, grid problems.	Instructor Notes
21	2	Classes & Objects	Practical: Create classes, objects, methods.	Zelle Ch.10
22	2	File Handling Concepts	Lecture: Open/read/write files, CSV structure.	McKinney Ch.3
23, 24	2	CSV Handling (Lab Program 5)	Practical: Read CSV, analyze contents.	Lab Manual 5
25	2	Exception Handling	Practical: try/except programs, custom exceptions.	Zelle Ch.11
26	2	Revision / Discussion		
27	3	Introduction to NumPy	Lecture: Arrays, indexing, slicing, shapes.	McKinney Ch.4
28, 29	3	NumPy Arrays (Lab Program 6A)	Practical: Create/reshape arrays.	Lab Manual 6
30, 31	3	NumPy Matrix Operations (Lab Program 6B)	Practical: Matrix addition, multiplication.	Lab Manual
32	3	Introduction to pandas	Lecture: Series, DataFrames, indexing.	McKinney Ch.5
33, 34	3	pandas DataFrame (Lab Program 7A)	Practical: Create DataFrame from CSV/dict.	Lab Manual 7
35, 36	3	Data Cleaning (Lab Program 7B)	Practical: Handle missing values, duplicates.	Lab Manual
37	3	Descriptive Statistics	Lecture: Mean, median, variance, SD, distribution.	McKinney Ch.7
38, 39	3	Descriptive Stats (Lab Program 8)	Practical: describe (), quantile (), var (), std ().	Lab Manual
40, 41	3	Data Distribution (Lab Program 9)	Practical: skew, kurtosis, value_counts.	Lab Manual
42, 43	3	Revision/Discussion		

Session	Module No.	Topic	Pedagogy /Activities	Reference for Study
44	4	Data Cleaning & Preprocessing	Lecture: Missing data, duplicates, IQR, outliers.	VanderPlas Ch.3
45, 46	4	Missing Data Handling	Practical: dropna(), fillna().	Lab Manual
47, 48	4	Categorical Encoding (Program 10)	Practical: LabelEncoder, OneHotEncoder.	Lab Manual 10
49	4	Scaling & Normalization	Lecture: Min-max, z-score, normalization.	McKinney Ch.7
50, 51	4	Scaling (Program 11)	Practical: StandardScaler, Normalizer.	Lab Manual 11
52	4	Outlier Theory	Lecture: IQR, visualization techniques.	Instructor Notes
53, 54	4	Outlier Detection (Program 12)	Practical: Boxplot, IQR detection.	Lab Manual 12
55	4	Advanced Pandas Operations	Lecture: merge, join, concat, groupBy.	McKinney Ch.8
56, 57	4	Advanced Pandas (Program 13)	Practical: merging, grouping, case study.	Lab Manual 13
58, 59		Revision / Discussion		
60	5	Introduction to Data Visualization	Lecture: Matplotlib basics, charts, customization.	McKinney Ch.9
61, 62	5	Matplotlib (Lab Program 12A)	Practical: line, bar, scatter, histogram.	Lab Manual
63	5	Seaborn Basics	Practical: boxplots, violin plots, pairplots.	VanderPlas Ch.4
64	5	Advanced Seaborn Visualization	Lecture: Themes, palettes, heatmaps.	VanderPlas Ch.4
65, 66	5	Correlation Heatmaps (Program 15)	Practical: Heatmap visualization for correlations.	Lab Manual
67	5	Plotly Introduction	Practical: Interactive plots.	Plotly Docs
68, 69	5	Identifying Trends	Practical: Line charts, moving averages.	Lab Manual
70, 71	5	Relationship Visualization	Practical: Scatter plots & regression lines.	Lab Manual
72, 73	5	Mini Project (EDA Case Study)	Practical: Real-world dataset analysis (finance/health/sports).	Instructor Notes
74, 75	1-5	Revision / Discussion		

Assessment and Evaluation

Continuous Internal Evaluations (CIE):

70 Marks

CIE	Mode	Pattern	Marks
CIE - 1	Quiz/Certification		20
Component – 1		Quiz (MCQ Based)	10
Component – 2		IBM Skills Build Courses: Graded	10
CIE – 2	Theory	Theory Examination	25
CIE – 3	Lab/ Practical		25
Component – 1		Lab Assignment Submission through GitHub/Google Classroom	05
Component – 2		Lab Project Work	20

Semester End Examination (SEE):

30 Marks

SEE	Mode	Pattern	Marks
	Practical	Lab / Practical Questions	30

Rubrics for evaluation of CIE (separate for each CIE component)

Assessment Component	Type of Component	Rubrics for Assessment	Refer Table for Rubrics
CIE 1	IBM Skills Build Courses	Certificate of completion for 10 marks. The specified courses on the IBM SkillsBuild platform to be uploaded to the designated Google Classroom.	Table (a)
	Quiz	Quiz will be conducted for 10 marks. Each right answer carries 0.5 mark each. No negative marking for wrong answers.	Table (b)
CIE 2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.	
CIE 3	Practical Component	Lab Assignment Submission through GitHub/Google Classroom for 5 Marks and Mini Project completion for 20 Marks	Table (c)
SEE	Practical Test	Successful Write up Completion of All Lab Questions: 10 Marks Viva-Voce: 05 Marks Lab Practical Test:10 Marks	Table (d)

Table (a):

Assess students' ability to complete assigned courses on the IBM SkillsBuild platform and reflect on their learning.

Criteria	Excellent (10 Marks)	Satisfactory (5 Marks)	Needs Improvement (0 Marks)
Completion of Courses (10 Marks)	Successfully completes all specified courses on the	Completes some courses or has minor errors in	Fails to complete the courses or does not upload certificates.

	IBM SkillsBuild platform before the deadline. • Certificates are uploaded to Google Classroom with no errors.	uploading certificates.	
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Table (b):

Criteria	Correct answer (0.5 Marks)	Wrong Answer (0 Marks)
Quiz questions (10 Marks)	Successfully choosing the correct option Student will be awarded (0.5 mark).	Fails to choose correct option will be reward as (0 mark).

Note: Total 20 Questions will be given Each right answer carries 0.5 mark. No negative marking for wrong answers

Table (c): Lab Assignment Submission through GitHub/Google Classroom

Criteria	Measuring Method	Excellent	Very Good	Good	Average	Needs Improvement
Code Quality & Logic (2)	Review of code in Google Classroom submission	Code is efficient, well-structured, uses correct logic; no errors (2)	Minor issues but overall correct and efficient (2)	Logic partially correct; minor errors (1)	Code runs with multiple errors; logic unclear (0.5)	Major issues; code does not run (0)
Output Accuracy (2)	Program execution and screenshots	Accurate output for all test cases (2)	Minor deviations in output (2)	Correct for basic cases only (1)	Output partially correct (0.5)	Output incorrect (0)
Timely Submission (1)	Timestamp in Google Classroom	Submitted before deadline (1)	Submitted on deadline (0.8)	Submitted within 12 hours after deadline (0.6)	Submitted within 24 hours after deadline (0.4)	Not submitted (0)

Table (c): Project Completion

Criteria	Excellent	Good	Poor
Implementation (5 Marks)	Identified a dataset, problem statement and effectively communicates the expected analysis 4-5 Marks	Identified a dataset and problem statement, communicates the expected analysis with minor areas of improvement 3-1 Marks	No review is provided 0 Mark

Report (5 Marks)	Report is well-written, organized, and thoroughly covers all necessary aspects with clarity. 4-5 Marks	Report is mostly clear and well-structured, with minor areas for improvement. 3-1 Marks	No report provided. 0 Mark
Presentation (5 Marks)	Engaging, clear, and well-structured presentation that effectively communicates key points. 4-5 Marks	Presentation is mostly clear and organized, with minor areas for improvement. 3-1 Marks	Presentation is missing or significantly incomplete. 0 Mark
Viva (5 Marks)	Demonstrates a profound understanding of the subject matter, answering questions with clarity and depth. 4-5 Marks	Answers questions accurately and confidently, with minor gaps in understanding. 3-1 Marks	Unable to assess subject knowledge due to incomplete or missing VIVA. 0 Mark

Table (d):

Criteria	Measuring Methods	Excellent	Good	Poor
Write-up (5 Marks)	Observations	Well-organized, clear, with proper logic of the problem. Validations with all test cases are handled. 5 Marks	Well-organized, clear, with partial logic of the problem and validations. 1-3 Marks	The student has not executed the program. 0 Mark
Execution and Format of Results (5 Marks)	Observations	Students demonstrate the execution of the program with error correction and show performance efficiency. Appropriate validations Execution Results should be show in proper format 5 Marks	Students demonstrate the execution of the program without error Correction and appropriate validations with a test case (minimum 1) are handled. Execution Results in without format 4 Marks	Execution Results with no comments and no output 0 Mark

Viva Voce Rubrics (Max: 5 marks)				
Criteria	Measuring Methods	Excellent	Good	Poor
Conceptual Understanding & Communication of Concepts (5 Marks)	Viva Voce	Explains thoroughly the program executed along with related concepts. (5 Marks)	Adequately explains the program executed along with related concepts (3 – 4 Marks)	Unable to explain the program executed along with related concepts (1-2 Marks)

Note: Overall lab tests will be conducted, including 2 programs where Write-up: 10 Marks & Viva-Voce: 5 Marks will sum up to 30 M.

Course Outcomes- Programme Outcomes Matrix

Program Outcome/ Course Outcome		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO1	Apply Python programming fundamentals and data structures to build basic programs, handle files, and organize data.	3	2	2	3	1	-	1	2	-	-	-	-
CO2	Implement data analysis using NumPy and Pandas, including descriptive statistics and data preprocessing techniques.	3	2	1	3	1	-	1	2	-	1	-	-
CO3	Create data visualizations using Matplotlib and Seaborn to	3	3	2	3	1	1	2	2	-	-	-	-

	represent data insights effectively.												
CO4	Develop and present a complete data analysis project demonstrating skills in programming, data manipulation, and visualization.	3	3	2	3	1	-	1	2	-	-	-	-

Dr. Mahadev K. Patil
Course Lead – Python Programming for Data Analytic